

Evidence-Centered Visualization for ML Security Engineering

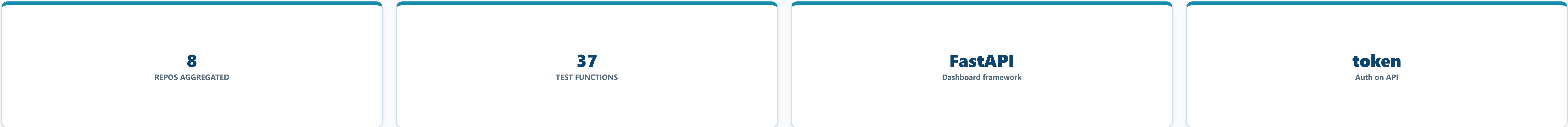
Aggregating Repository-Level Measurements Into Traceable Technical Dashboards

github.com/poojakira/mlsec-dashboards (public, default branch main, primary language Python). MIT • Python 3.12 • source snapshot b02a0fc • verified 2026-10-01

EVIDENCE POSTURE

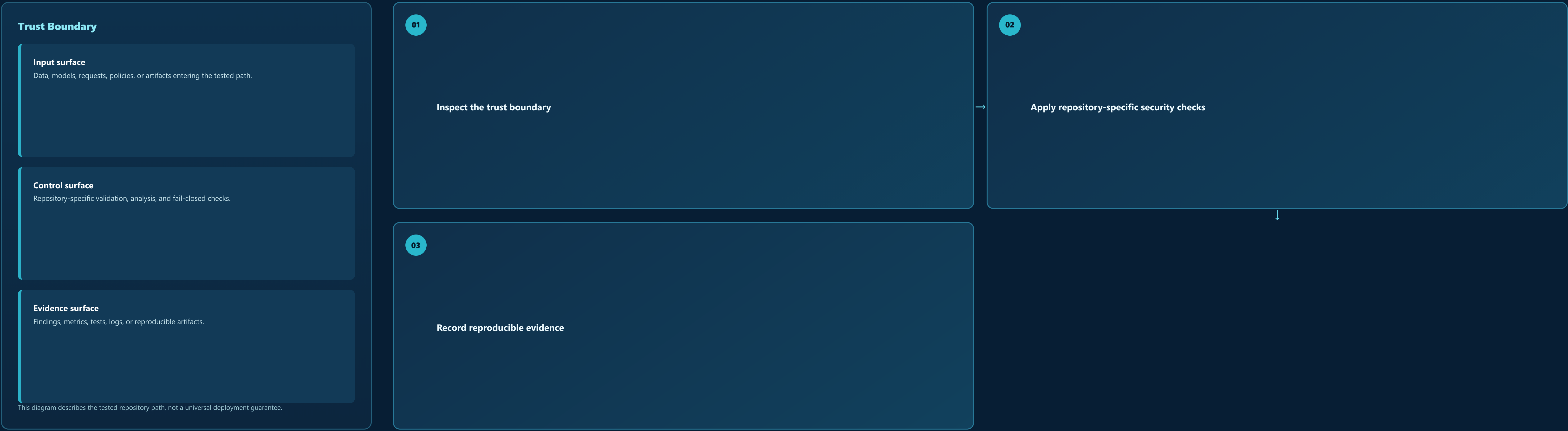
Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.



SYSTEM VIEW

Architecture and Threat-Control Flow



Research Contribution

An evidence-centered dashboard server that aggregates committed per-repo JSON into traceable HTML + a token-authed REST API, deliberately surfacing weak results (e.g. ROC-AUC 0.54, OOD F1 0.72) alongside strong ones rather than curating highlights.

Verified Evidence

EV-01	VERIFIED_AT_SNAPSHOT - Aggregates evidence from 8 active ML security repos - README: dashboard_server.py discovers sibling evidence/*.json.	EV-02	VERIFIED_AT_SNAPSHOT - 37 test functions - Counted def test_ definitions in tests/ at source snapshot b02a0fc; current README also records 37 passing pytest tests.
EV-03	VERIFIED_AT_SNAPSHOT - Shows weak results explicitly (OOD F1 0.72, ROC-AUC 0.54) - README states weak- alongside-strong design; numbers sourced from sibling evidence files.	EV-04	VERIFIED_AT_SNAPSHOT - Token-authed REST API + static HTML - README features; dashboard_server.py.
EV-05	UNSUPPORTED (disclaimed) - Live SOC monitoring / real-time alerting - README: static evidence snapshots, not live monitoring. Not claimed.	EV-06	Design system: shared

Method and Control Path

- 01Inspect the trust boundary
- 02Apply repository-specific security checks
- 03Record reproducible evidence

Limitations and Residual Risk

- Static evidence snapshots, not live monitoring.
- Data can lag the current CI state of each repo.
- No alerting or real-time telemetry.
- Depends on sibling repos committing evidence.
- Local aggregation; not a hardened service.

Reproducibility

Evidence: dashboard_server.py, sibling evidence/*.json, tests/

Evidence Provenance

- Claim discipline**
Metrics come from the repository's verified metrics and research brief.
- Scope discipline**
Benchmarks are reported only at their documented fixture, dataset, or test scope.
- Reproduction**
PDF and PNG are generated locally from committed evidence without paid services.